

INTERMEDIATE ALGEBRA

Chapters 1 & 2 Quiz

Basic Techniques for Solving Equations • Functions Review

Name: _____ Date: _____ Score: _____ / 100

Directions. Ten questions, 10 points each. Show all of your work — a correct answer with no supporting work earns little credit, and a wrong answer with sound work earns partial credit. Simplify all answers, and give domains and ranges in interval notation. No calculators.

Part I — Chapter 1

Basic Techniques for Solving Equations

1. Solve the following inequality. Write your answer in interval notation.

$$x - 1 \leq 3x + 2 \leq 2x + 6$$

2. Solve the following system:

$$4x + 5y = 43,$$

$$9x - 2y = 57.$$

3. Jeff is 4 times older than his daughter. Five years ago he was 9 times older than his daughter. How old is his daughter?

4. Find all ordered pairs (x, y) that satisfy both

$$\sqrt{x} + \sqrt{y} = 7 \quad \text{and} \quad 3\sqrt{x} - 4\sqrt{y} = -14.$$

5. Find all solutions to the following system of equations:

$$\begin{aligned} 8x + y - z &= 46, \\ 3x + 4y - 2z &= 27, \\ 4x - y + z &= 14. \end{aligned}$$

Part II — Chapter 2

Functions Review

6. Find the domain and range of

$$S(x) = \frac{12x - 9}{6 - 9x}.$$

7. Find the domain of

$$f(x) = \frac{1}{\sqrt{2x - 5}} + \sqrt{9 - 3x}.$$

8. Suppose that a function $f(x)$ is defined for all real x . How can you use the graph of $f(x)$ to produce the graph of $y = |f(x)|$? Explain why your method works.

9. Let $f(x) = \frac{4x}{x+2}$ and $g(x) = \frac{2x}{x+4}$. Find $f(g(x))$.

10. The function f defined by

$$f(x) = \frac{cx}{2x+3}$$

satisfies $f(f(x)) = x$ for all real numbers x except $-3/2$. Find c .